Business Statistics

Variables and Scales of Measurement

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Variables and Scales of Measurement Overview

- Variables
  - Independent & dependent
  - Explanatory & response
  - Characteristics of data

- Scales of Measurement
  - Nominal
  - Ordinal
  - Interval
  - Ratio

Variables



Variables and functions
When a value can vary

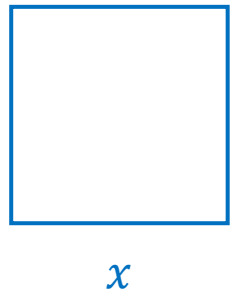
- Recall that a function takes inputs and generates a single output.
- The idea of a function is that there is a fixed quantitative relationship between the inputs and the output
- As the relationship holds even when the input values "vary," the inputs of the function are **variables** and are written with letters like x
- Similarly, the output "varies" depending on the inputs, and so the output is also a **variable** and are written with letters like y
- Statistics use variables in different contexts, but at the end of the day, the variables are letters or words that represent a value that can vary

Variables
Independent and dependent variables

- A **function** represents a **relationship** between different values
- Consider a square which has 4 sides of equal length
- The perimeter equals the sum of the lengths of all the sides
- If the length of a side is x, and the perimeter is y, then

$$y = 4 * x$$

- Picking a value for x automatically determines the value for y
- As we can freely pick a value for x, it is an **independent** variable
- As the value of y "depends" on x, then y is a **dependent** variable



Functions
Variable and function naming and usage

- The perimeter of a rectangle equals twice the width plus twice the height.

perimeter = 2 * width + 2 * height

- This can also be written as a function showing the input variables.

$$perimeter = f(width, height)$$

- Width and height are independent variables which can be varied. They can also be identified by letters.

$$X_1 = width$$

$$X_2 = height$$

- Perimeter is a dependent variable, as its value depends on the values selected for width and height, again written as a letter.

$$Y = perimeter$$

- A function then is the relationship between the independent and dependent variables

$$Y = f(X_1, X_2)$$

Variables
What varies in experiments?

- *Does taking multi-vitamins reduce cognitive decline in older adults?*
- An **explanatory** variable "explains" or influences another variable
- A **response** variable "responds" to changes in the explanatory variable
- Randomized control experiments are used to test **hypotheses**, and include a control group and an experimental group
- The experimental group is given a **treatment**, which is the condition or level of the explanatory variable
- The control group is either not given a treatment, or given a placebo
- How much the response variable changes is used to reject or fail to reject the hypothesis

Variables
Characteristics in the data set

- For our data sets, the **variables** are the **characteristics** in our data
- Our small data set has 4 subjects and 2 characteristics per subject
- There are therefore 2 variables:
  - Location and Sales in May

Store ID	Location	Sales In May
A	Downtown	42,000
B	Downtown	35,000
C	Airport	58,000
D	Campus	31,000

- These 2 characteristics are variables because their values "vary" depending on which subject is observed
- The sales for Store A are 42,000, but for Store D they are 31,000
- The location value for Store B is Downtown and for Store C is Airport

Variables

Categorical or Numerical

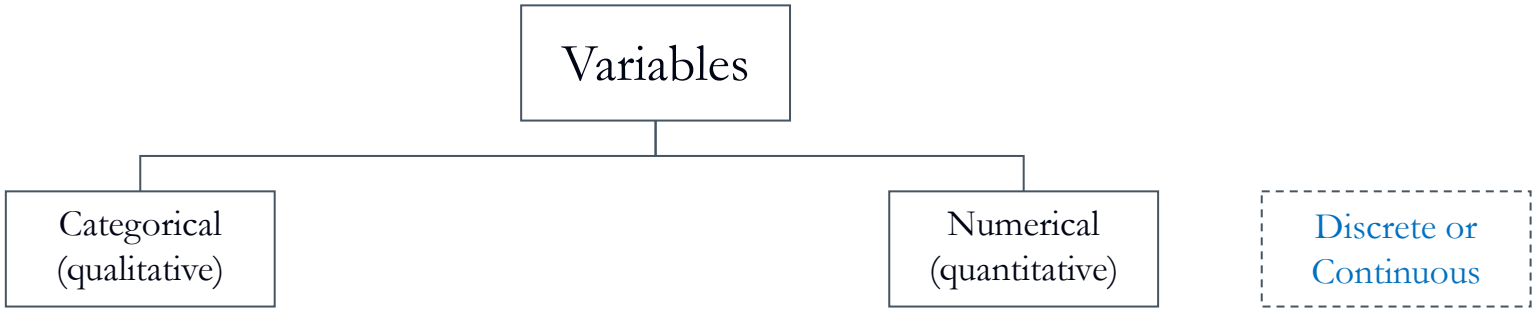


Variables
Classified as either categorical or numerical

- **Numerical variables**
  - Represent a numerical value (5, -6, 2.59, 42,000, etc.)
  - They are **quantitative** - "quantities"
  - Can either be discrete or continuous
  - Represent meaningful numbers (**not** numbers used as merely labels)
- **Categorical variables**
  - They do not represent meaningful numbers
  - They are **qualitative** - "qualities"
  - Represent **categories**

Store ID	Location	Sales In May
A	Downtown	42,000
B	Downtown	35,000
C	Airport	58,000
D	Campus	31,000

Scale of Measurement
Variables are either categorical or numerical



Discrete: countable, separate numbers
Number of credit card transactions, number of website clicks

Continuous: a value within a range
Student GPA, persons' height, time between website clicks

Scales of Measurement

Scale of Measurement
Variables cannot take on just any value

- Variables are limited to the type of value they can take
- Those types are defined by their scale of measurement

  - Nominal
  - Ordinal
  - Interval
  - Ratio

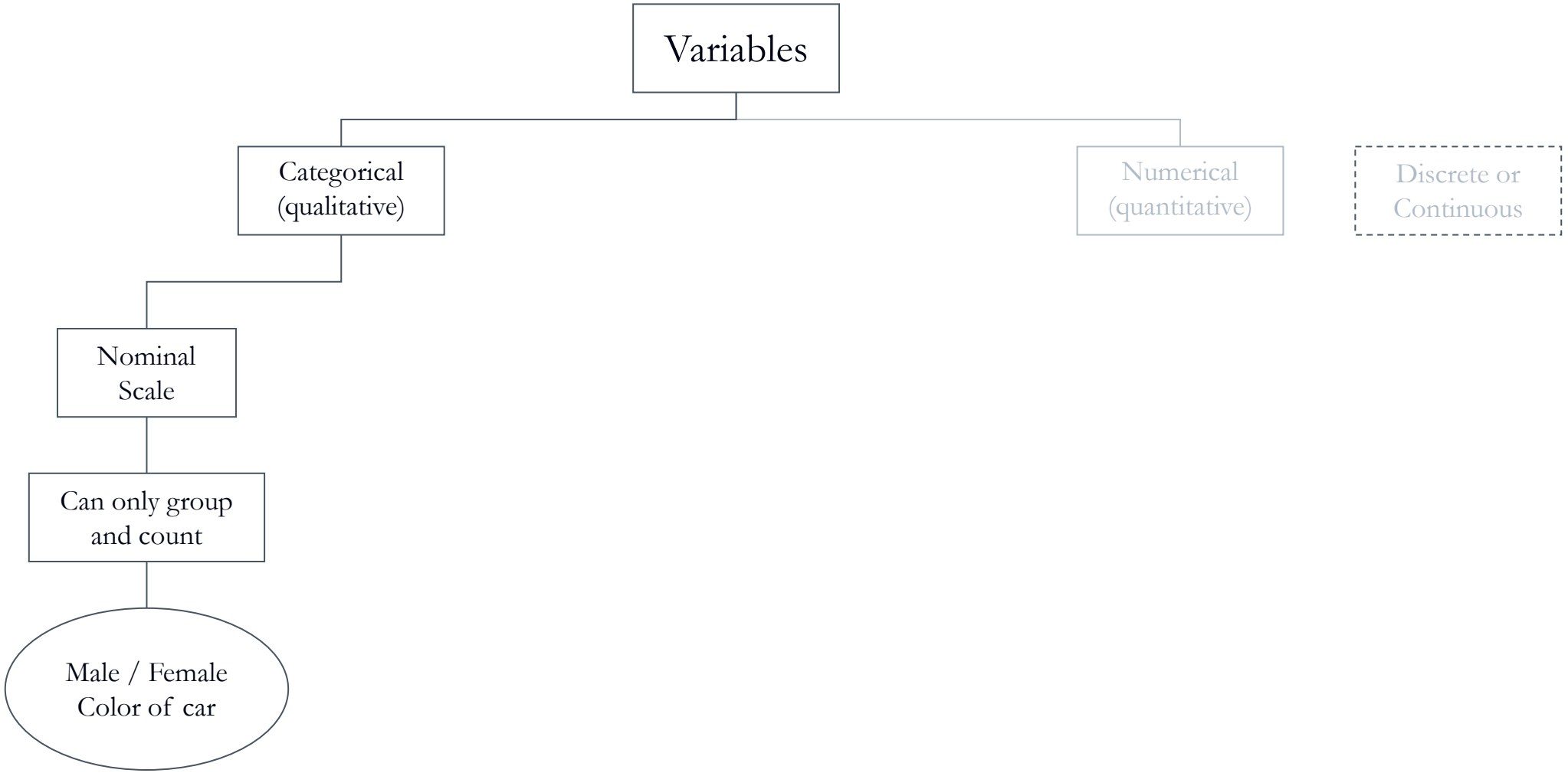
StoreID	Store Type	Customer Satisfaction	Store Open Time (AM)	Monthly Sales
101	Urban	High	8	48,500
102	Suburban	Medium	9	39,200
103	Rural	Low	10	27,400
104	Urban	High	8	51,800
105	Suburban	Medium	9	40,100

Scale of Measurement
Nominal scale - Categorical variables

- Observations are labels, even if they are coded with numbers
  - For example, if Urban = 1, Suburban = 2, and Rural = 3
  - **Does not** mean that 2 Urban locations = 1 Suburban location
- Nominal variables can only be grouped and counted

StoreID	Store Type	Customer Satisfaction	Store Open Time (AM)	Monthly Sales
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Scale of Measurement
Nominal observations are categorical variables

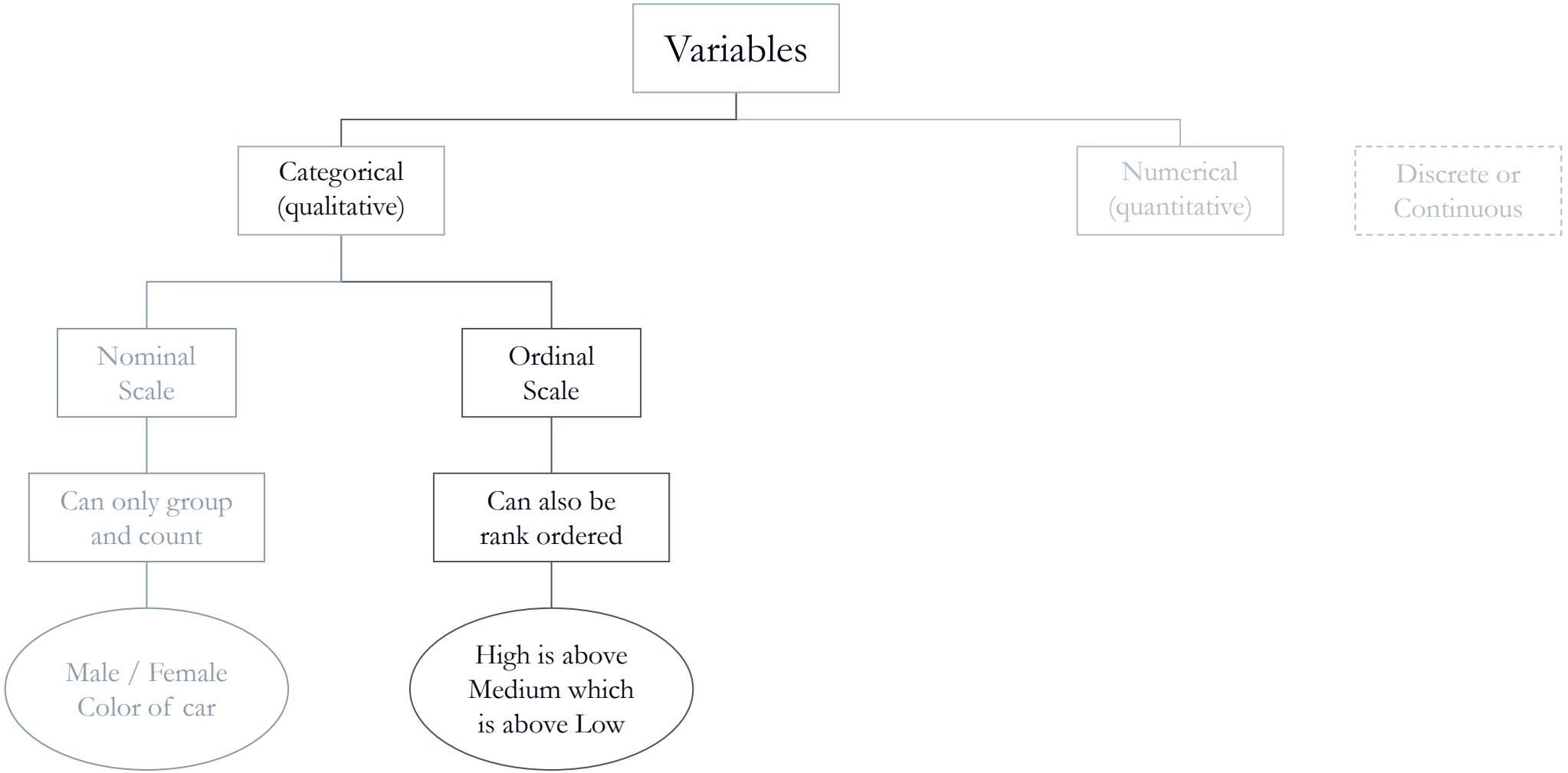


Scale of Measurement
Ordinal scale - Categorical variables

- Ordinal variables have some "order"
  - Even if numbers are used, their actual value is meaningless
  - A survey score of 5 could be either "strongly agree" or "strongly disagree"
- Ordinal variables can be grouped and "ordered" or ranked

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Scale of Measurement
Ordinal observations are categorical variables

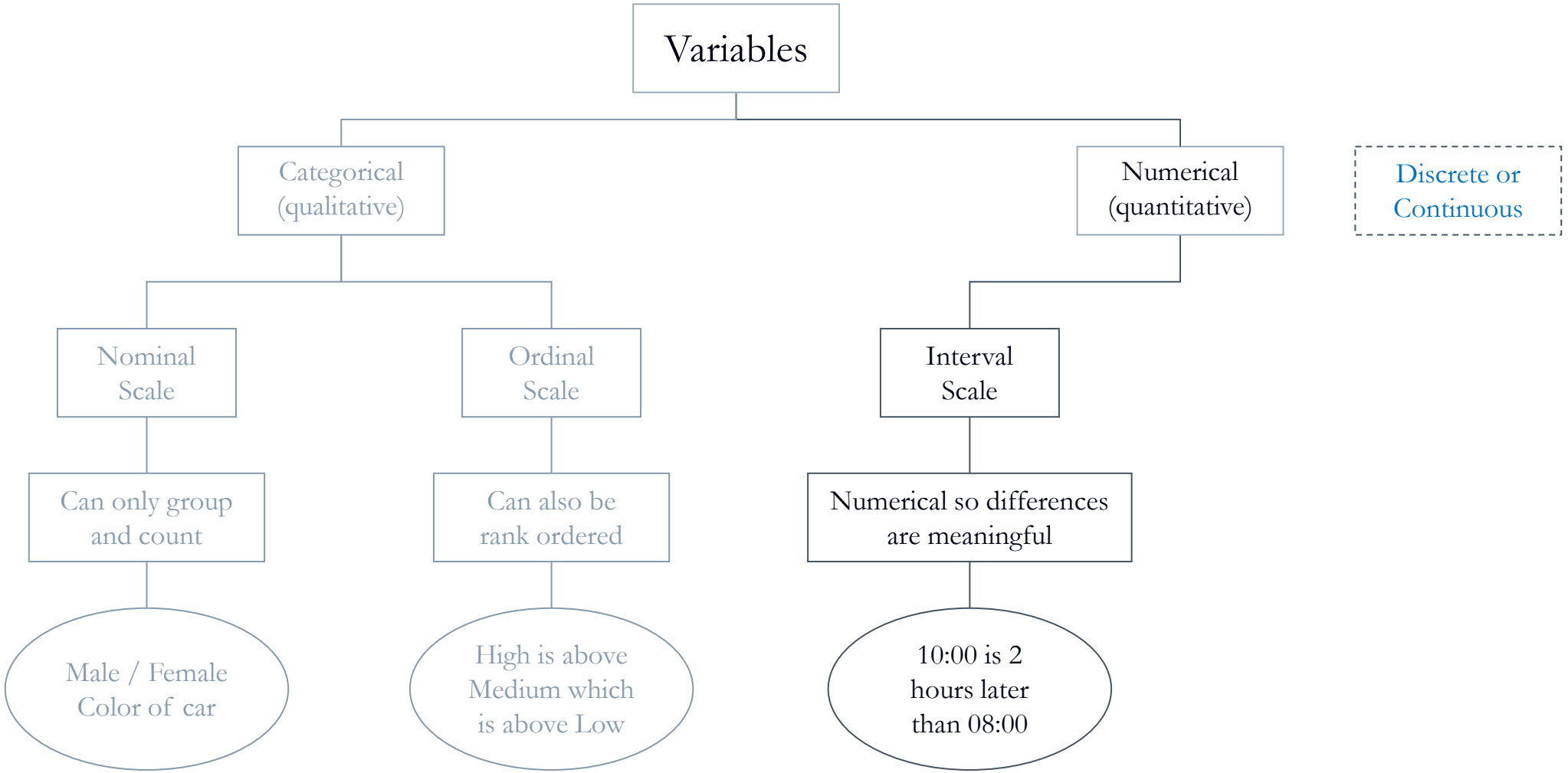


Scale of Measurement
Interval scale - Numerical variables

- Can perform some arithmetic operations
- Observations have a meaningful difference between their values
  - 10:00 is 2 hours later than 8:00
- Without a fixed zero point though, ratios are meaningless
  - **Cannot** say that 2 * 5:00 = 10:00

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Scale of Measurement
Observations on an interval scale are numerical variables

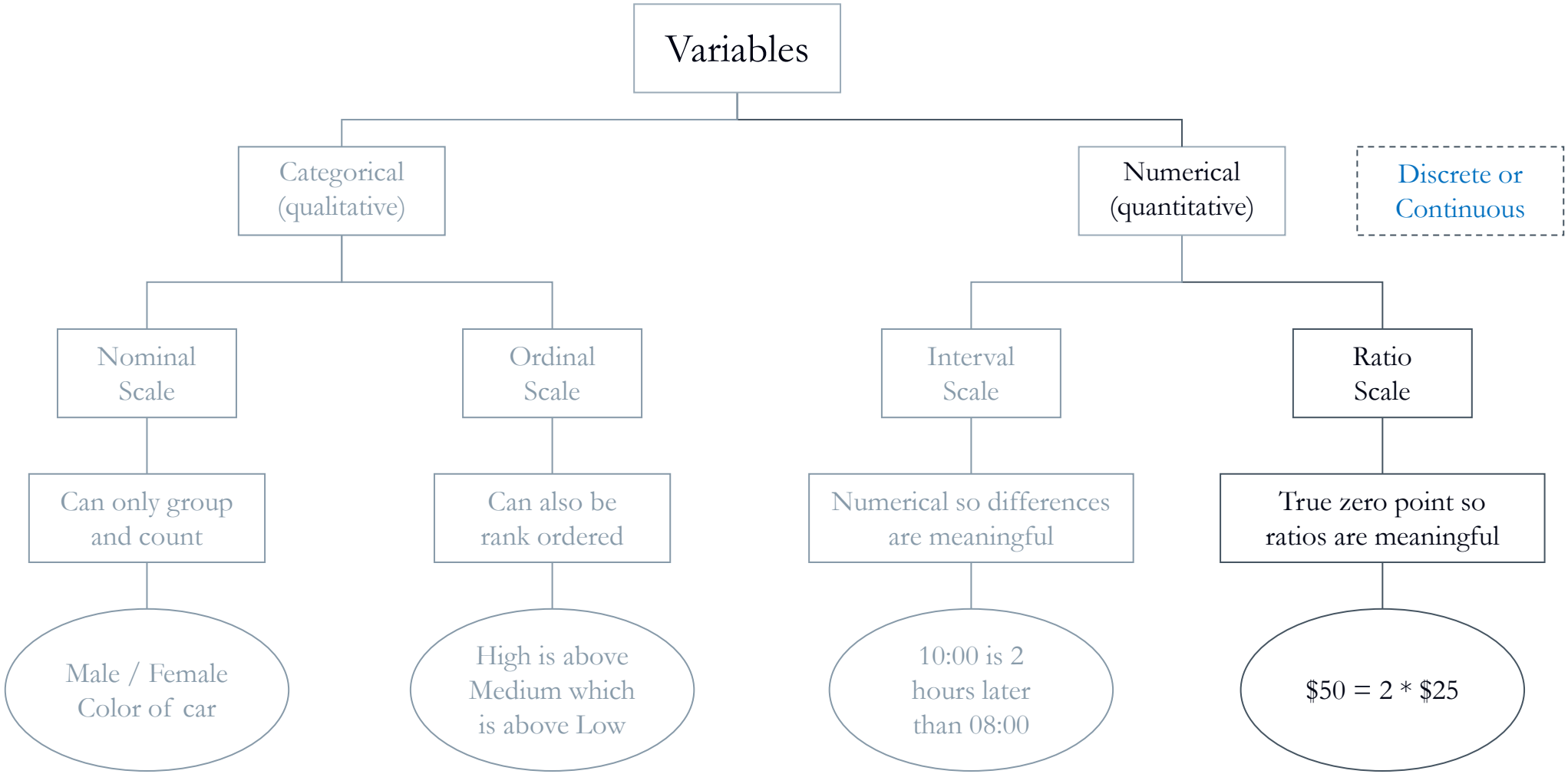


Scale of Measurement
Ratio scale - Numerical variables

- Can perform arithmetic operations
- Observations have a meaningful difference between their values
  - $51,800 is $3,300 greater than $48,500
- With a true zero point, ratios are now meaningful
  - $50,000 is twice as much as $25,000

StoreID	Store Type	Customer Satisfaction	Store Open Time (AM)	Monthly Sales
101	Urban	High	8	48,500
102	Suburban	Medium	9	39,200
103	Rural	Low	10	27,400
104	Urban	High	8	51,800
105	Suburban	Medium	9	40,100

Scale of Measurement
Observations on a ratio scale are numerical variables



Scale of Measurement
Variable types are based on their measurement scale

